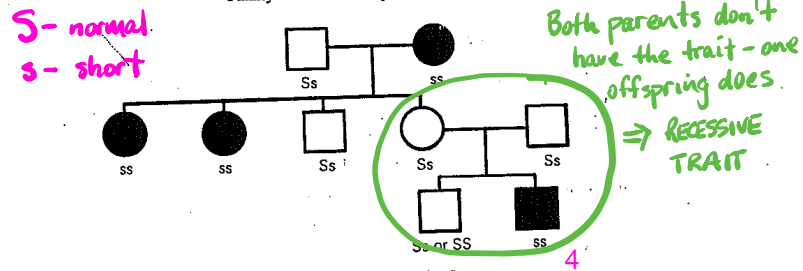


Exercise 11.1 Family Trees

INTRODUCTION:
When we draw family trees the squares represent males and the circles females. Individuals that show the character under investigation are shaded in.
Carefully study the following family tree that shows the incidence of short eyelashes.

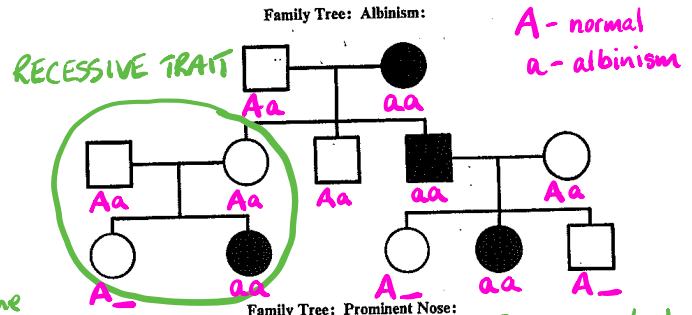
Family Tree—Short Eyelashes



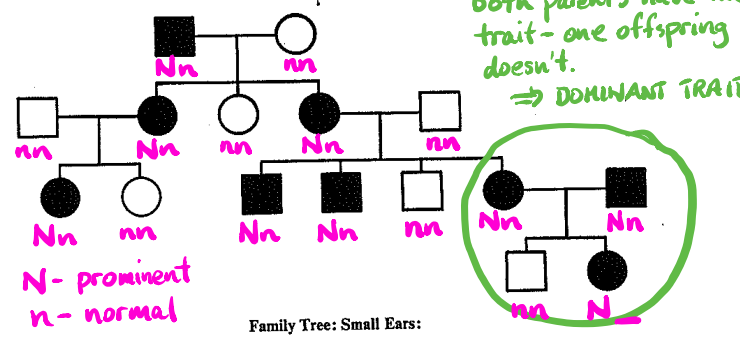
- (i) How many members of this family have short eyelashes?
How do you know?4 coloured shapes
- (ii) How many members have long eyelashes?5
In the family tree the genotypes are indicated below the squares and circles.
- (iii) Do the shaded individuals have homozygous or heterozygous genotypes?
Homozygous
- (iv) Is short eyelashes a recessive or dominant character?
Recessive
- (v) One male member of the family tree has two possible genotypes. Explain
He has not had children yet so the second allele cannot be determined yet.

Procedure:
Carefully study each of the family trees, For each one you are to work out if the character is dominant or recessive and indicate the genotype of each number.

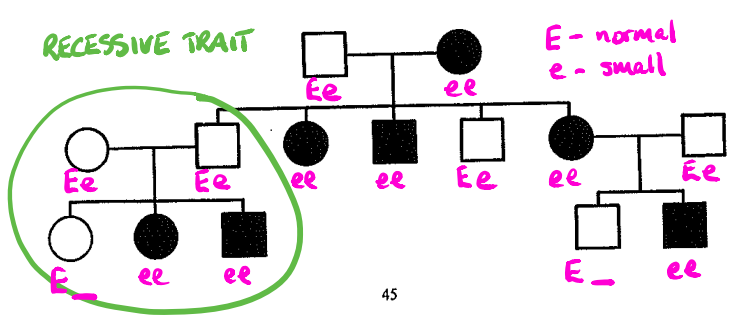
Family Tree: Albinism:



Family Tree: Prominent Nose:



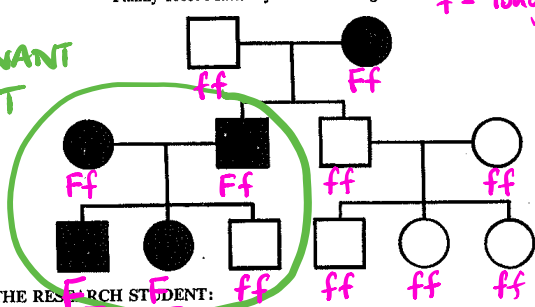
Family Tree: Small Ears:



DOMINANT TRAIT

Family Tree: Markedly Shortened Fingers:

F - short
f - long



FOR THE RESEARCH STUDENT:

A boy has red hair (homozygous recessive) and neither of his parents has red hair but his father's father had red hair. Draw up a possible family tree to show how the boy inherited his red hair.

BIOLOGICAL TERMS TO LEARN: family tree

RESEARCH REPORT:

Exercise 12.1 The Ability To Taste PTC— Dominant or Recessive?

INTRODUCTION:

PTC refers to the chemical phenylthiourea. Your teacher will have soaked filter paper in the prescribed amount of PTC to make the PTC papers required in this exercise. In this exercise you are to determine whether the ability to taste PTC is due to a dominant or recessive gene.

PRACTICAL WORK:

Materials: (per student)
distilled water
PTC papers (3 cms by 1 cm)

Procedure:

Wash out your mouth with distilled water (if available) to eliminate any conflicting tastes that might be present.

Place one PTC paper in your mouth and chew on it. PTC has quite a strong taste so you will definitely know if you can taste it or not. Record your result by ticking the appropriate box.

I am a:— PTC taster ☐
PTC non-taster ☐

Remove the PTC paper from your mouth and place it in the bins provided....

Total up the number of tasters and non-tasters in your class. Record the results by writing the number in each box. If available total up other class results.

Class results	PTC tasters	
	PTC non-tasters	
Combined results of several classes	PTC tasters	
	PTC non-tasters	

- (1) How did "tasters" describe the taste of PTC?
- (2) Why could some students taste the PTC while others could not?
- (3) What percentage of your class were tasters?